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Reasoning and Acting (ReAct) With Multimodal LLMs: A Framework for Intent Driven 6G Networks

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ABSTRACT 6G wireless networks are poised to realise Self-X capabilities through the native integration of an AI-plane within the network architecture. This AI-plane will enable autonomous orchestration and management driven by user intent. Recent advances in Generative AI (GenAI) introduce powerful capabilities for such intent-based management, particularly through the representation of multimodal data in a shared embedding space. This enables the development of autonomous agents that not only translate intent but also manage network functions autonomously, both before and after deployment. In this paper, we propose a comprehensive framework for reasoning-and-acting (ReAct) agents powered by multimodal large language models (LLMs), facilitating seamless translation of user intent into network actions. The proposed architecture integrates multimodal retrieval-augmented generation (RAG), persona-driven reasoning, and function calling to orchestrate tasks across vision, text, and radio modalities. We examine how RAG influences reasoning in multimodal agents, how LLMs can be specialised for telecom applications, and which performance benchmarks should guide their design. A UAV-based coverage optimisation case study demonstrates the framework's ability to leverage vision data to enhance average signal-to-noise ratio (SINR), yielding substantial reductions in outage probability and improvements in throughput under Rician fading channels. Analytical characterisation and practical implementation results are presented to benchmark performance.

INDEX TERMS 6G, LLM, GenAI, UAV, multimodal agents, RAG, agentic architecture, reasoning.

I. INTRODUCTION

A. MOTIVATION

6G WIRELESS networks have already started to take shape within 3GPP [1], [2], [3], [4], with research and development (R&D) geared to design, test, and showcase initial deployments [5] by 2030. Key requirements [6], [7] for 6G have also been identified by the operators, requiring increased focus on:

- 1) Supporting new services, e.g., Metaverse [8], Integrated Sensing and Compute (ISAC) [9], and AI-on-RAN (inferencing as a service) [10];
- 2) Monetisation through exposure of APIs, e.g., CAMARA [11], [12], and new services, e.g., through integration of non-terrestrial networks (NTN) [13];
- 3) Energy efficiency and spectrum efficiency through incorporation of AI-native approaches [14];
- 4) Simplification of the network and backward compatibility, e.g., through simplification of the control plane [15] and user plane functions and use of Multi-RAT spectrum sharing (MRSS) [16].
- 5) Improved service with ubiquitous connectivity, e.g., through direct-to-device [17] connectivity.

To that end, the architecture for 6G networks should ensure resilience, security, reliability, sustainability, efficiency, interoperability, and programmability from day one while improving the customer quality-of-experience (QoE). To meet the design requirements, often network design needs to strike a balance across various trade-offs. For instance, for the increased utilisation, the network should allow load balancing across sites by evenly distributing the user equipments (UEs), while on the other hand, for potential energy savings, the network should unevenly distribute/offload the UEs across sites so some of them can enter sleep mode [18]. With the large design space, taming the complexity to ensure optimal performance manually is almost impossible. Consequently, almost all architectural discussion for 6G considers induction of AI-plane [19], [20] with increased virtualisation and disaggregation to manage the complexity.

The AI-native 6G RAN should allow zero-touch deployment, optimising the operation decision based on *intent*. The *intent-based management* [21], [22], [23] of the network separates expression and management of operational requirements from how these can be met through service orchestration and management solutions. This allows implementation of 6G RAN as an autonomous system (as specified by the TM Forum Reference Architecture [24]). However, translation of the intent into dynamic service level objectives (SLOs) is a non-trivial task even when the services have fixed ontological descriptions [25]. Interpreting intents within different contexts is challenging, e.g., a high-level business intent can be broken down into customer intent and subsequently resource management intent [21].

Large language models (LLMs) provide tools which can be easily harnessed to provide contextual understanding. The application of LLMs for intent based management is already gaining momentum [26], [27], [28]. In [26], the authors employ LLM to process user intents and extract key details, such as the target performance metric and the desired optimization level. Subsequently, a Transformer-based intent validation mechanism is applied to identify and discard intents that conflict with the current network state, thereby preventing potential performance degradation. The proposed approach outperforms the baseline method, achieving improvements of 12% in throughput, 26.5% in delay reduction, and 17% in energy efficiency. In [27], authors employ LLM for intent parsing, as well as code generation for radio resource management. As shown in the seminal paper [29], LLMs can also be used to implement autonomous agents, transitioning from language models to large action models (LAMs) [30]. The reasoning and acting (ReAct) framework can exploit chain-of-thought (CoT) [31] or tree-of-thought [32] prompting to develop LLM reasoning capabilities based on text. In [33], authors have explored how CoT can be applied to wireless networks and what are the key challenges. These advancements have ultimately led to the development of LLM-based agents capable of not only understanding the intent but also translating it into AI/ML

or classical optimisation based workflow which can optimise the networks autonomously.

Agentic AI offers a proactive approach to network management, surpassing traditional AI/ML methods that are largely reactive, as noted by GSMA [34]. While a few conceptual studies have explored agentic AI in wireless networks (e.g., [35] and others summarized in Table 1), there remains a lack of comprehensive research, considering the multimodal nature of RAN data. In particular, the combined consideration of multimodality and the reasoning capabilities of agents, where their interplay can pave the way toward truly Self-X networks is yet to be investigated. To that end, this contribution is geared to address this research gap.

B. RESEARCH GAPS AND CONTRIBUTIONS

When deploying multimodal LLM-powered agents, the following design issues need to be addressed:

- 1) The Reasoning and Acting (ReAct) framework requires formal treatment to clarify why reasoning actions do not modify the environment state, and how function calling should be constrained accordingly. For wireless networks, where multiple algorithms can yield equivalent solutions at different Pareto-optimal operating points, this motivates the identification and formalisation of a stochastic generalisation of ReAct.
- 2) Employing multimodal agents for wireless network RRM inherently couples the performance of the foundation model with network-level outcomes. Key performance measures—including latency, function-call relevance detection, and energy efficiency—must be formally defined, as they are mutually coupled and directly influence how quickly the network can be re-programmed through agentic reasoning.
- 3) Comprehensive treatment of multimodality within agentic AI for wireless networks remains lacking. Reasoning over multimodal data can be realised either through embodied models with shared embedding spaces or through modality transformation (e.g., vision-to-text). Both approaches, and their implications for ReAct-based agents, need to be explored.
- 4) Agent personas—which define the role and behavioural context of an agent—are typically assigned arbitrarily in wireless network literature. Understanding how persona design influences reasoning trajectories and task performance is an open question that requires attention.
- 5) For agentic implementations, it is important to understand how in-context learning through retrieval-augmented generation (RAG) influences orchestration capabilities. This includes whether agents can learn from previously completed task trajectories and compose new methodologies based on semantic similarity.
- 6) A generic multimodal framework for agents must address the architecture for agent memory, planning, and the toolsets relevant to wireless network RRM tasks.

TABLE 1. Comparison of This Paper and Other GenAI for Telecom Studies.

Framework/Article	Unique contribution	Modalities	Key Metric/Findings	Use Cases / Tasks	Architecture & Agent Design
This article	Comprehensive framework for multimodal agents with tool orchestration and reasoning. Implementation demonstrated with a UAV-based coverage optimisation case study.	Text, Image, Radio	Reduction in outage probability and higher mean throughput. The framework also identifies latency, relevance, and energy efficiency as key criteria for foundation-model selection.	UAV-assisted coverage optimisation and intent-driven wireless network orchestration.	ReAct + function calling, multimodal RAG, memory, personas, and Gemini & SAM2 integration.
Mohsin et al. [36]	RAG based LLM for exploiting multimodal sensing information for wireless propagation.	Image and Object Features	RAG-based LLM optimization for context-aware pipeline achieves 8%, 8%, 10%, 7%, and 12% improvements in relevance, faithfulness, completeness, similarity, and accuracy over baseline LLMs.	Understanding the wireless environment through multimodal sensing data.	RAG pipeline with unified vector DB and multimodal embeddings.
AI2MMUM (Jiao et al. [37])	Unified foundational model for multimodal telecom data for enabling several down-stream tasks. Foundation models for IQ and waveform design explored in [38] and [39] also.	RF time series, Text	Accuracy on telecom physical-layer tasks is shown to be comparable to the state-of-the-art.	LOS/NLOS, beam selection, path loss (DeepMIMO, WAIRD) several down-stream tasks on same foundation model.	Telecom LLM backbone + radio encoders + LoRA fine-tuning + task heads.
ENWAR (Nazar et al. [40])	ENWAR integrates multimodal sensing to perceive, interpret and plan complex task in dynamic wireless environment.	DeepSense dataset, along with textual, image and time-series data	Environmental awareness through multimodal sensing. ENWAR shows improvement over vanilla LLMs.	Multimodal sensing.	Multimodal RAG agent with sensor fusion of GPS, LiDAR and camera.
LAMBO (Dong et al. [41])	Large AI model-based offloading framework for mobile edge computing using transformer-based encoder-decoder architecture.	Task metadata + system state	Achieves lower latency and energy consumption compared to traditional deep offloading methods.	Task offloading and resource allocation in MEC.	Asymmetric encoder-decoder with actor-critic pre-training and active learning from expert feedback.
WirelessGPT (Yang et al. [42])	Generative pre-trained foundation model for multi-task wireless communication and sensing.	RF time series (CSI, channel data)	Demonstrates competitive performance on channel estimation, prediction, and human activity recognition tasks.	Channel estimation, prediction, and environmental reconstruction.	Transformer-based foundation model with unsupervised pre-training and task-specific fine-tuning.
Wang et al. [43]	LLM based approach for base-station siting. Another study on network optimisation is also presented in [44].	Language + Spatial data or encoded few-shot optimisation examples	Demonstrated distinct advantages, ranging from reducing human intervention to enabling highly automated and adaptive solutions.	LLM agents for telecom infra planning.	Prompt-tuned LLM agent for base station siting using RAG.
Tang et al. [45]	Autonomous composition of Edge AI services using LLMs and the code generation capability.	Language + metadata	Demonstrated the usage using hugging face and integration of meta-data processing pipeline.	Edge AI orchestration (e.g., slice, RIC).	LLM agent as rApp for O-RAN orchestration with AI services.
Lee et al. [46]	The paper studies the LLM based optimisation for wireless networks. Instead of using black-box approaches the LLM optimisation is modeled as Markov process.	Text + historical actions	The study proves the convergence of LLM based optimisation for wireless networks.	Network optimisation.	LLM-based optimizer as finite-state Markov chain (LLMO).
WiLLM (Liu et al. [47])	WiLLM is open source framework for enabling LLM service over 5G wireless network. The framework expands open-air interface (OAI) ecosystem to enable enhanced UE capabilities.	Textual prompts + network telemetry	The study presents dual-layer slicing, app-layer tunneling, cross-layer APIs.	LLM-aware wireless comm for smart glasses.	LLM-tailored wireless RAN core network ecosystem.
WirelessAgent (Tong et al. [48])	Modular agentic LLM for autonomous wireless network control with task-specific agents and coordination mechanisms.	Text + Network state metadata	The study demonstrates that the agent-based approach provides 44% gain compared to prompt-based approach.	Task-specific network optimisation and control.	LLM agent (perception, memory, planning, action) using LangGraph.
Dev et al. [35]	Conceptual framework for integrating agentic AI in 6G networks.	Network data	Framework for network management using AI agents.	Network optimisation.	Advanced 6G architecture with control/user plane integration + agentic AI for real-time learning, energy, and resource optimization.

7) Lastly, practical implementation of such a framework on both real and synthetic data is needed to demonstrate viability for real-world deployment.

In this paper, we present a comprehensive treatment of these design issues. Through a case study deploying the proposed multimodal agentic framework for UAV-assisted communications in a rural scenario, we demonstrate practical viability. The case study incorporates dynamically generated functions and exploits multimodal LLM capabilities in place of classical algorithms—for example, using the LLM’s visual reasoning to determine the number of clusters K for K -means clustering, replacing the need for algorithmic determination.

We present analytical characterisation under Rician fading channels and show that the agentic implementation substantially reduces outage probability and increases throughput for all UEs. To the best of our knowledge, this paper presents one of the first comprehensive treatments of a multimodal agentic reasoning framework for optimisation of wireless networks.

C. ORGANISATION

The rest of this article is organised as follows. Section I-D provides an overview of research in the area of specialising LLMs for telecom networks. In Section II, a formal treatment of the ReAct framework is presented, including performance measures for selecting foundational LLMs (Subsection II-B),

multimodal agents (Subsection II-C), and the role of personas (Subsection II-D). Building on these elements, Section III develops a comprehensive framework for intent-based management of wireless networks using multimodal agents. Section IV provides a case study in which the framework is implemented for a UAV coverage-optimisation scenario, presenting implementation details, performance analysis, and key results. Section V presents future work and research directions. Section VI outlines the risks and limitations associated with deploying agentic AI in telecom systems. Lastly, Section VII presents the conclusion.

D. RELATED WORK AND BACKGROUND ON LLMs FOR TELECOM

Agentic AI implementations in telecom networks rely heavily on the underlying foundational LLM. These models can either be general-purpose or specialised for telecom applications through pre-training, fine-tuning, or in-context learning. Among these, retrieval-augmented generation (RAG) has emerged as a practical approach for incorporating domain-specific knowledge without incurring significant training overhead.

Several recent works have explored the use of LLMs in telecom scenarios. In [49], the authors develop a telecom-specific LLM through continual pre-training and instruction tuning, demonstrating improved performance on telecom-oriented tasks such as modelling, Q&A, and coding. Similarly, [50] proposes domain-adapted LLMs using curated telecom corpora. These approaches highlight the benefits of domain-specific adaptation but require significant computational resources and curated datasets. RAG-based approaches provide a lighter-weight alternative: in [51] and [52], RAG pipelines are employed to build telecom-specific assistants capable of interpreting standardisation documents, with enhancements such as semantic chunking [53] further improving retrieval quality. However, these approaches are primarily designed for text-based interaction and do not consider multimodal data or agentic reasoning.

Recent studies have begun exploring multimodal and agentic AI in wireless networks. Mohsin et al. [36] investigate RAG pipelines for multimodal sensing data, while AI2MMUM [37] proposes unified foundation models for RF time-series and textual data. Frameworks such as ENWAR [40] and WirelessAgent [48] demonstrate the feasibility of LLM agents for network optimisation through task-specific agent design and coordination mechanisms. LLM-driven optimisation has also been explored for infrastructure planning [43], [44], convergence analysis [46], and O-RAN orchestration [45], [47]. In addition, recent works such as LAMBO [41] have explored large AI model empowered edge intelligence, while WirelessGPT [42] proposes a generative pre-trained multi-task learning framework for wireless communication. These studies collectively advance the integration of LLMs into wireless network control loops.

Despite these advances, most existing approaches exhibit one or more of the following limitations: (i) they focus

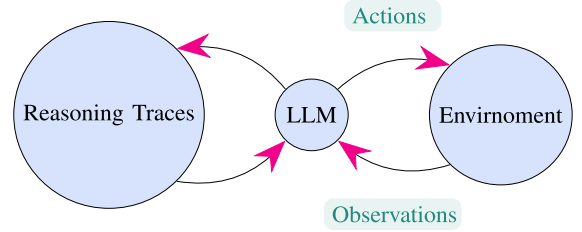


FIGURE 1. Reasoning and acting framework.

on unimodal data, primarily text; (ii) they address specific downstream tasks without providing a unified treatment of reasoning and tool orchestration; or (iii) they rely on static pipelines or prompt-based optimisation without explicitly modelling the reasoning-action loop required for autonomous network management. In contrast, the framework proposed in this paper introduces a unified multimodal agentic architecture that integrates ReAct, multimodal RAG, and function calling within a single system. Unlike prior works, the proposed framework explicitly accounts for the interactions among multimodal data (vision, text, and radio), agent personas, and tool orchestration, while providing a comprehensive treatment of performance metrics (latency, relevance detection, and energy efficiency) that couple LLM behaviour with network performance. Table 1 provides a detailed comparison with recent related studies.

II. REASONING AND ACTING (REACT) EMPOWERED MULTIMODAL AGENTS

A. INTRODUCTION TO REACT AND FUNCTION CALLING

ReAct [29] exploits the capability of LLMs to generate both reasoning and task-specific action traces in an interleaved manner. In other words, agents¹ try to accomplish the overall task by iterating through thought → action → observation sequence (see Fig. 1). More formally,

ReAct Framework

Consider an arbitrary agent $x \in \mathcal{X}$ interacting with a certain environment $e \in \mathcal{E}$ to accomplish task $t \in \mathcal{T}$, where $\mathcal{X}, \mathcal{E}, \mathcal{T}$ are the collection of agents, environment families, and set of tasks to be completed by x . The environment e may transition through states which can be monitored through a series of *observations*. So, at time t , an agent receives observation $o_t \in \mathcal{O}$ from the environment and performs action $a_t \in \mathcal{A}$ following some policy $\pi(a_t|c_t)$, where $c_t = (o_i, a_i, o_i \forall i = 1, 2, \dots, (t-1))$ is the *context* provided to the agent. Learning such a policy is challenging and requires extensive computation [29]. To this end, ReAct augments x 's action space to create $\bar{\mathcal{A}} = \mathcal{A} \cup \mathcal{L}$ where $\mathcal{L}$ is the space of language-based reasoning. An action $\bar{a}_t \in \mathcal{L}$ is referred to as a *reasoning* action or a *thought*. The reasoning or thought does not influence the state of the environment, therefore it has no external impact. However, it aims to create

¹In the context of this paper, an agent can be considered as an object which has both attributes, toolsets, and also possesses the capability to synthesise tools, e.g., through code generation.

useful information by reasoning over the current context c_t , providing a way to update it to $c_{t+1} = (c_t, f(\bar{a}_t))$ where $f(\cdot)$ is some function invoked as a result of the thought. The thoughts and reasoning keep the agent's actions grounded. This prevents unnecessary drift or hallucinations.

ReAct framework is the basis of many other agent implementation frameworks which extend the basic capability by enhancing the toolset. The ReAct plan–orient–act cycle allows decomposition of tasks into sub-tasks. Nevertheless, it does not provide the LLM with a way to return structured responses for particular tasks. This can be accomplished by pairing the ReAct agent with what is known as “*function calling*” [54]. Function calling equips the LLM with a toolset that it can invoke by inferring the context of the prompt. By employing ReAct and function calling in a synergistic manner, the LLM can then *chain* appropriate tools to accomplish complex tasks.

Function Calling & Stochastic Generalisation

Consider an arbitrary agent $x \in \mathcal{X}$ and an associated current context $c_{t+1} = (c_t, f(\bar{a}_t))$ as described above, then the function calling is effectively a way of selecting action a_{t+1} , i.e., $c_{t+1} \rightarrow a_{t+1}$. Notice that the function calling can adopt probabilistic selection by associating a probability of the form $p(a_{t+1}|c_{t+1})$. In other words, the LLM invokes a certain action with a certain probability under a given policy π . However, in most of the LLM based agent implementation literature, this probabilistic model has not been explored yet. Most literature deals with deterministic mapping of toolsets based on reasoning to solve a particular task. The probabilistic nature is particularly important for the tasks where two or more functions can accomplish the same task albeit yielding a different solution optimality and thus operational point on the Pareto frontier.

In wireless networks, the stochastic formulation is particularly relevant: different optimisation strategies (e.g., clustering, scheduling, or power allocation) may achieve similar objectives but at different points on the performance–complexity trade-off. By associating a probability distribution $p(a_{t+1}|c_{t+1})$ over candidate actions, the agent can explore this trade-off and adapt based on context or environmental conditions, providing robustness in dynamic scenarios where deterministic policies may fail to generalise. As a concrete illustration, consider a UAV agent choosing between K -means clustering and DBSCAN for user grouping. Under deterministic function calling, only one method is invoked based on the reasoning trace. Under stochastic generalisation, the agent assigns probabilities to each method based on contextual cues—such as observed user density uniformity—enabling adaptive tool selection that explores different Pareto-optimal operating points without requiring explicit rule-based switching logic.

In summary, function calling enables LLM agents to tap into domain-specific knowledge bases and to outsource tasks to established toolchains (e.g., language interpreters, optimisation solvers). Recent efforts have produced open

foundation models specialising in function calling, such as GRANITE-20B-FUNCTIONCALLING [55]; interested readers are referred to [55] for further details.

B. PERFORMANCE MEASURES & KEY DESIGN CONSIDERATIONS

Berkeley's Function-Calling Leaderboard (BFCL) [56] provides a foundation for evaluating function-calling capabilities across LLMs, considering task diversity, programming languages, and multi-step invocation. While comprehensive benchmarking is beyond the scope of this paper, several performance measures are critical when designing agents for wireless networking tasks, as the behaviour of the underlying LLM is tightly coupled with network optimisation outcomes.

Relevance Detection

In practical deployments, an agent must handle diverse tasks and interact with multiple toolchains. It is therefore essential for the agent to determine whether invoking a function is necessary for a given intent. Function relevance detection quantifies the agent's ability to avoid unnecessary or incorrect function calls, and can be evaluated by benchmarking against scenarios where no function invocation is required. The expected behaviour is that the agent refrains from calling any functions, thereby avoiding hallucinated actions or incorrect parameter usage.

Besides relevance detection, latency is another critical metric. Deploying the LLM on an edge device may reduce invocation delay but increase inference latency due to limited compute; distributing computation across an edge–cloud continuum can balance these factors. Therefore, latency must be analysed at both the model and system levels.

Latency

Latency is defined as the time taken by an LLM to invoke an appropriate function based on a given prompt, excluding pre-processing of the function specification. In practice, it can be measured as token generation rate (tokens per second) or as end-to-end delay for function invocation.

Cost is another important consideration, serving as a proxy for energy consumption and computational complexity. Following BFCL [56], the cost associated with function calling can be expressed as

$$\mathfrak{C} = \frac{\sum_{i=1}^N \Delta_i}{N} \times \frac{M \times C}{T}, \quad (1)$$

where N is the number of function calls, Δ_i is the latency of the i^{th} call, M is the number of GPUs used during inference, C is the cost per GPU per hour, and $T = 3600$ is a normalisation factor. This formulation can be extended to estimate carbon emissions by mapping C to energy consumption metrics provided by cloud platforms [57].

Energy Consumption

Let $x \in \mathcal{X}$ be an agent executing a policy $\pi \in \Pi$ to accomplish a task $t \in \mathcal{T}$, where π consists of actions a_i , each realised through N_i function calls. The average energy

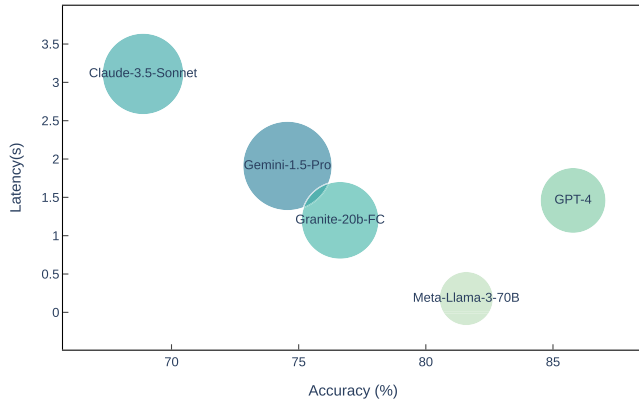


FIGURE 2. Average Latency vs. Average accuracy for function calling of different models.

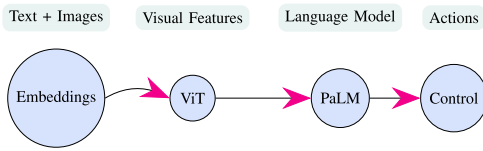


FIGURE 3. PALM-E architecture.

consumption is:

$$E_{Model} = \sum_{i=1}^{||\Pi||} \sum_{j=1}^{N_i} \Delta_j \zeta, \quad (2)$$

where $||\Pi||$ is the number of actions, Δ_j is the latency of the j^{th} function call, and ζ is the normalised hardware-dependent energy consumption per unit time. In practice, ζ captures the average power draw of the compute platform (e.g., GPU or edge device) during inference, and can be estimated via hardware profiling tools or cloud provider energy specifications.

These metrics provide a baseline for comparing foundation models for ReAct-based agent implementation. Fig. 2 illustrates the trade-off between latency and accuracy across different models, with bubble size indicating relevance detection performance. In practice, these metrics can be obtained through controlled benchmarking of inference latency, function-call accuracy, and hardware-level energy profiling under representative workloads.

C. MULTIMODAL AGENTS

The rise of multimodal LLMs naturally provides a platform for development of multimodal agents. These foundation models form the core of such agentic implementation. Nevertheless, the task of implementing reasoning and acting within multimodal agents has only been investigated recently in [58] and [59]. There are two different approaches for utilising multimodal LLMs to create multimodal agents.

1) EMBODIED MULTIMODAL LANGUAGE MODEL

PaLM-E [59] is an embodied multimodal LLM that injects embodied observations such as vision data, state space

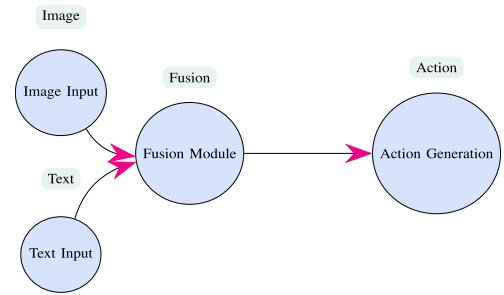


FIGURE 4. MM-ReAct.

estimates, and sensor streams into the language embeddings of a pre-trained LLM. The model encodes continuous multimodal data from sensors into vector embeddings with the same dimension as the underlying LLM. The output in textual format represents decisions or plans, which can then be fed into a model that translates high-level plans into low-level control for robotic applications. While embodied models inject continuous data representations directly into the LLM's embedding space, this assumes that the embedding space is rich enough to represent world models. Fig. 3 shows the architecture of PaLM-E.

2) FUNCTION CALLING BASED MULTIMODAL REACT

The second approach relies upon an extension of ReAct through the injection of multimodal observations, termed MM-ReAct [58]. MM-ReAct is training-free and does not require an encoder for cross-modal embeddings. Instead, it relies on function calling to develop tools that serve as vision experts. The outcome of these experts can subsequently be used to generate thoughts in the reasoning process, guiding the behaviour of agents in accomplishing their intended goals. Fig. 4 depicts the architecture of MM-ReAct. Due to the simplicity and the inherent advantage of not requiring re-training, we consider this approach as a potential candidate for implementing multimodal agents in future wireless networks.

The performance of multimodal agents can be quantified using the metrics introduced in Section II-B. However, additional factors—including persona design and the integration of radio-frequency (RF) data alongside vision and text modalities—also influence the agent's reasoning behaviour. While the case study in this paper primarily employs vision and text, the proposed framework is inherently extensible to RF data such as channel state information (CSI) and spectrograms, which can be integrated via specialised tools within the function-calling pipeline. The fusion strategies for incorporating such heterogeneous modalities are discussed in detail in Section II-E, and the role of personas is addressed next. Importantly, the function-calling architecture is modality-agnostic: RF-specific tools, such as CSI estimators, spectrogram classifiers, or time-series forecasters, can be registered as additional callable functions without any architectural modifications to the agent or the ReAct loop.

D. PERSONAS

The agent *persona* defines its role within a wider ecosystem. For instance, within a RAN deployment, multiple agents may exist with roles such as RF optimiser, resource allocation manager, or orchestrator. Persona essentially forms the soul of an agent [60], giving it a distinct personality. While personification is most visibly important in user-facing applications (e.g., chat agents), the question of whether personas impact an agent's *reasoning* capabilities is equally relevant for network optimisation tasks.

Initial evidence from [61] demonstrates that assigning personas improves reasoning capabilities compared to standard zero-shot prompting. However, personas must be carefully designed, as they can introduce intrinsic biases [62] and, if misconfigured, can degrade reasoning performance [63]. To mitigate this, [63] proposes a framework that combines persona-based and neutral prompting, passing outputs through multi-shot reasoning steps to select the best outcome. In this paper, we assume such safeguards are absorbed into the persona definition process.

For wireless network applications, effective persona construction should reflect the agent's functional role by explicitly encoding task objectives, available toolsets, and operational constraints. In the UAV coverage optimisation case study presented in Section IV, the persona is defined as a "flying base station" with specific capabilities (onboard camera, satellite backhaul) and a clear goal (identify users, compute centroids, calculate path-loss and achievable rates). This structured persona definition constrains the agent's reasoning to the relevant action space and prevents the invocation of irrelevant tools. A systematic ablation study evaluating the impact of different persona configurations—such as varying the level of specificity in goal descriptions, changing the assigned functional role, or comparing persona-based versus neutral prompting—on task execution outcomes, including throughput, function-call relevance, and reasoning chain length, is an important research direction. Such an evaluation requires controlled experimentation across multiple agent instances and tasks, and is therefore left as dedicated future work.

E. RAG AND MULTIMODAL RAG

In addition to personas, the agent's behaviour can be refined by tuning the underlying LLM through low-rank adaptation (LoRA) [64], retrieval-augmented generation (RAG) [65], or a combination of pre-training, instruction, and alignment tuning [49]. These techniques differ significantly in training cost, energy efficiency, and accuracy. In this paper, we focus solely on in-context learning via RAG and explore how the traditional RAG pipeline can be expanded to embrace multimodality.

Specialising LLMs on telecom standards through RAG has been explored in [51], [52], [53], and [66]. For agentic implementation, we focus on two key design questions: (1) How does RAG affect the agent's ability to orchestrate functions based on intent? (2) How can RAG be extended

to support multimodal data for specialising the underlying LLM?

1) AGENTIC RAG

Recent studies [67], [68] demonstrate that an agent's reasoning capabilities are strongly influenced by the architecture for domain-specific information retrieval. In [67], the authors show that providing task-specific context—distilled from raw environment state via the LLM—improves orchestration of function calls. They further propose using exemplar memory, where past task trajectories are retrieved via semantic similarity search on task-specific metadata, to guide the LLM's planning process.

However, [68] identifies key limitations of this approach: semantic similarity between tasks does not guarantee similar solutions; injecting full trajectories suffers from context length constraints; and historic trajectories may contain irrelevant steps that degrade orchestration. To address these issues, the authors propose the Thought Retrieval and Aligned Decision (TRAD) framework, which retrieves demonstration steps based on *generated thoughts* rather than raw task metadata, and temporally aligns retrieved sequences with the current task state to improve action prediction. The detailed steps of TRAD are presented in Appendix B.

In summary, context-specific information retrieval demonstrably enhances agent orchestration capabilities. Therefore, telco-specific agent implementations should employ domain-specific RAG—preferably within a TRAD-type framework where retrieval directly supports reasoning—on either a generic or telco-specific fine-tuned LLM.

2) MULTIMODAL RAG AND GENERAL ARCHITECTURE

Most existing RAG pipelines are unimodal and primarily text-based. A truly multimodal agent should leverage multimodal data during information retrieval. The core idea behind multimodal RAG is to represent all modalities in a common embedding space, enabling joint retrieval across text, vision, and other data types. In [69], the authors demonstrate the superiority of multimodal RAG over traditional text-only pipelines, though comprehensive evaluation of generality remains open.

For telecom applications, multimodality extends beyond visual data to include time-series data.² LLMs have been shown capable of analysing and forecasting time-series data [71], and techniques such as cross-modal alignment fine-tuning (CALF) [72] are being developed for time-series alignment with LLMs. Based on these considerations, we propose a unified framework for multimodal agentic RAG, shown in Fig. 5, which provides the foundation for the agentic scaffolding detailed in the next section.

²The need for LLMs to operate on time-series data and the development of wireless foundation models has been discussed in [70].

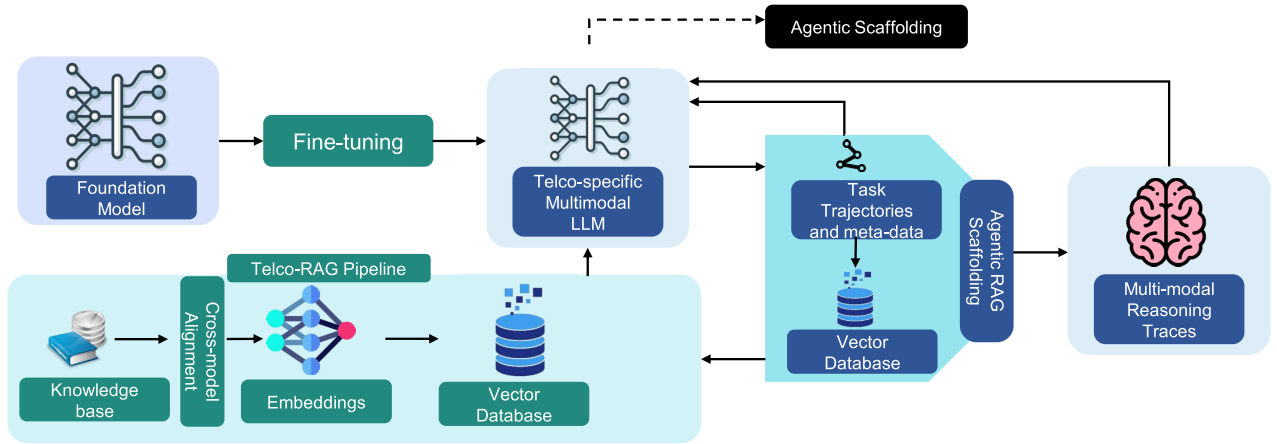


FIGURE 5. General architecture for the multimodal Agentic RAG framework.

3) FUSION STRATEGIES FOR MULTIMODAL DATA IN 6G NETWORKS

In practical 6G deployments, multimodal data arises from heterogeneous sources: textual intent descriptions, visual data (e.g., camera feeds), and RF measurements (e.g., time-series signals, CSI, spectrograms). Incorporating these into a unified RAG framework requires appropriate fusion strategies.

Early fusion projects embeddings from different modalities into a shared latent space before retrieval, enabling joint semantic similarity search across modalities. This allows reasoning over correlated information (e.g., associating visually detected user locations with RF propagation characteristics), but requires careful embedding alignment. *Late fusion* processes each modality independently and combines results at a higher decision level—more modular but potentially limiting cross-modal interactions. A *hybrid* approach selectively fuses modality-specific representations depending on the task, which is particularly relevant when RF data requires specialised encoders while text is handled directly by the LLM.

In the proposed framework, we adopt a function-calling based multimodal integration strategy, where modality-specific tools (e.g., vision models, RF estimators) are invoked dynamically within the ReAct loop. This avoids strict embedding alignment while enabling flexible, task-driven fusion—well-suited for dynamic wireless environments where the relevance of different modalities varies over time and across network conditions.

III. PROPOSED FRAMEWORK

Building on the ReAct framework, performance metrics, and design factors introduced in Section II, this section weaves these elements into a cohesive framework for intent-based management of wireless networks. A subset of elements is validated through the case study in Section IV.

A. MULTIMODAL LLM EMPOWERED AGENTS

The fundamental element of any agentic implementation is its capability to understand both “intent” and “goal” from

natural language. By invoking the underlying LLM, agents can reason to infer task-specific intent from the prompt. To prevent injection of malicious intents, these must be regulated through guard-rails [73]. The agent has *five* key architectural blocks: i) memory; ii) planning pipeline; iii) domain-specific LLM fine-tuning; iv) toolset and functions for performing tasks; and v) action orchestration mechanism.

1) MEMORY

The agent memory can be classified following the cognitive architecture for language agents (CoALA) [74], which divides memory into three types:

- 1) **Procedural Memory:** The working memory containing the fine-tuned LLM code and agent code, updated only when the foundational model or agent code changes.
- 2) **Semantic Memory:** The long-term knowledge base, stored as embeddings in vector stores (see Fig. 6). This includes the agent’s persona and domain knowledge, retrieved and embedded into prompts for LLM interaction.
- 3) **Episodic Memory:** Records of past actions that guide the agent’s future task execution.

The agent implementation can reside on edge or cloud infrastructure, with memory distributed across the edge–cloud continuum. Placing semantic and episodic memory close to the agent improves inferencing throughput; the RAG cache proposed in [75] reduces time-to-first-token by up to 4× and improves throughput by up to 2.1×. For multimodal agents with time-series processing capabilities, streaming data stores such as KafkaML [76] can be integrated into semantic memory. For cognitive wireless networks, the memory store should also accommodate the radio network information base (RNIB), control/user plane analytics, and service delivery information [77].

2) PLANNING

Planning is a core operation that distinguishes agentic implementations from simple function chaining. The ability to reason about tool relevance and sequence invocations

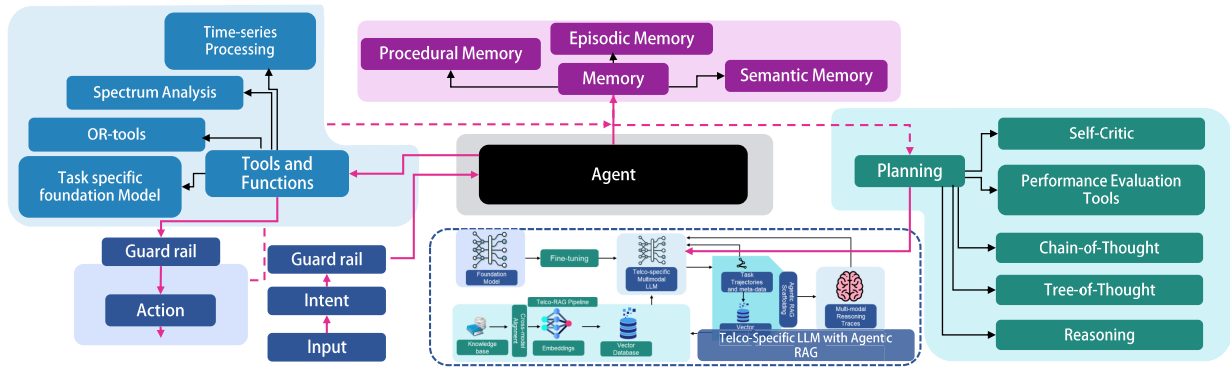


FIGURE 6. Overall agent implementation framework.

according to the presented intent is essential. Our proposed framework combines ReAct and TRAD (see Section II-E) as shown in Fig. 6: ReAct forms the core reasoning loop, while TRAD is employed for complex multi-step task planning with domain specificity.

Both frameworks interleave chain-of-thought (CoT) prompting [31] with action-plan generation. However, CoT's sequential nature can underperform in tasks where good initial decisions are crucial. Tree-of-Thoughts (ToT) [32] generalises CoT by considering multiple reasoning paths and self-evaluating choices, naturally enabling exploration for optimal decision making. The number of look-ahead steps and search algorithms are design knobs that can be configured per task [78]. Whether to perform CoT through discrete or continuous embedding vectors remains a matter of active debate [79].

The choice of planning strategy depends on the deployment scenario. Tasks requiring code generation demand precise planning where poor initial decisions cascade into faulty implementations. In contrast, RAN data analysis tasks may benefit from ToT's exploration capability to weigh different approaches when confronted with unforeseen data patterns.

3) TOOLS AND FUNCTIONS

To execute the inferred plan, the agent requires tools and functions that can be invoked. Function definitions and semantic descriptions are provided to the LLM as context, enabling incorporation into the planning process. Tool implementation can follow a microservice-based architecture distributed across the edge-cloud continuum. For 6G wireless network optimisation, the multimodal agent can possess multiple toolsets, including:

- 1) **Time-series analysis:** Network performance data stored as key performance metrics (KPMs) can be analysed using cross-modal embeddings or dedicated time-series foundation models such as TimesFM [80], which achieves zero-shot performance comparable to supervised approaches.

- 2) **Spectrum analysis:** Identification of incumbents and dynamic adjustment of co-existence parameters can be realised through FFT-based spectrogram analysis or transformer-based waveform classification. Foundation models for spectral analysis of 6G radio are emerging [81], while image-based approaches for waveform classification are more mature [82].
- 3) **OR tools:** Standard operations research tools (e.g., Google OR Tools) for knapsack, travelling salesman, and vehicle routing problems can be exposed as microservices for the agent.

Task-specific tools may also be required for particular scenarios. The lifecycle and dependency management of these tools forms a central part of AgentOps. It is also possible to implement task-specific tools through a code generation pipeline employing in-context learning via traditional or Graph RAG. In our case study, we demonstrate how tool generation is possible by invoking the same foundational model through prompting.

IV. MULTIMODAL AGENT BASED FRAMEWORK FOR UAV COVERAGE OPTIMIZATION

In this section, we provide a detailed case study demonstrating how the core elements of the proposed framework can be implemented in practice. The case study covers practical design issues for real-world implementation and demonstrates that the agentic framework can yield significant performance gains.

A. SCENARIO AND INTENT DEFINITION

We consider a UAV platform equipped with a camera and two transceivers. One transceiver is used for command-and-control communication over a backhaul link, as well as for downlink data transmission from the ground station via satellite link (e.g., Starlink or OneWeb).³ The UAV's camera reference frame is aligned with the flying platform via a gimbal. The UAV transmits in the N77 band [85] within

³Without loss of generality, data can also be relayed via high altitude platforms for multi-tier non-terrestrial networks (NTNs) [83], [84]. However, our performance analysis focuses on the access side, so this assumption does not impact the discussion.

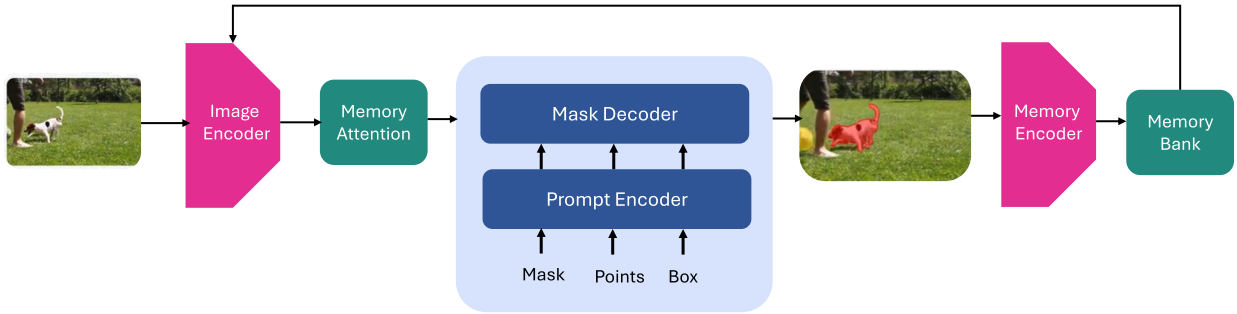


FIGURE 7. SAM2 implementation model and key components.

the 3.3–4.2 GHz frequency range, where transmissions are primarily influenced by path-loss. The agent on the UAV is tasked to track users and position the platform optimally to maximise achievable downlink rates on the Aerial-to-Ground (A2G) channel. We quantify gains using a snapshot approach, i.e., a semi-static crowd tracking scenario⁴ where throughput gains are measured by optimally positioning the UAV. Our focus is on demonstrating the agentic behaviour rather than solving the optimal scheduling problem itself; readers interested in resource allocation are directed to [86]. The persona, intent, and methodology are captured by a single prompt:

Initialisation of Agent and Goal]
We initialise an agent by selecting an appropriate foundation model and providing it with a persona, intent, and goal in a single prompt.

Instruction ← You are a flying base station. You have an onboard camera. You use the camera to find the users in field of view. You then hover over to the user and provide them downlink connectivity. You are back-hauled on the Star Link satellite connection. Your goal is to identify the users in the field of view. You then have to calculate centroid of the location of users. You then need to calculate the path-loss between centroid, i.e. your hovering location and the users. You then use this to calculate the achievable downlink rates. You cannot receive any input from me apart from your location which is provided as GPS coordinates.

Notice that the highlighted elements represent the persona, intent, methodology, and goal. We do not specify which tools should be invoked; the agent decides this dynamically. These tools can also be powered by the same foundation model used to implement the agent, as demonstrated in this case study.

B. IMPLEMENTATION DETAILS

1) CHOICE OF PLATFORM AND HARDWARE

The proof-of-concept is implemented in Vertex AI Studio[®] hosted on Google Cloud, running a managed Jupyter Notebook instance configured as summarised in Table 2. We utilise the open dataset of UAV images from [87].⁵ This implementation demonstrates agent capabilities with-

TABLE 2. Configuration for Managed Vertex AI Studio Instance.

Element No:	Components:	Configuration:
1	VM Instance Type	n1-standard-2
2	vCPUs	2 vCPUs with Intel Skylake
3	Memory	7.5 GB
4	CPU Architecture	x86
5	GPU	NVIDIA T4
6	GPU Memory	16 GB
7	Disk Size	150 GB
8	Hosting Region	europe-west4-a
9	Models used	Gemini 1.5 & SAM2

```
1 #Initialise Agent and configure safety settings
2
3 model = GenerativeModel("gemini-1.5-pro-001",
4                           system_instruction=[Instruction],
5                           generation_config={..}, safety_settings=[..
6                           ],, )
7
8 #Notice that Instruction must be initialised as
9   per the prompt presented in previous
10  discussion, generation configuration
11  includes initialisation model temperature
12  and safety setting can be set as per the
13  requirements.
```

LISTING 1. Agent initialisation.

```
1 {
2   "name": string,
3   "description": string,
4   "parameters": {object (Schema)}
5 }
```

LISTING 2. Function specification.

out focusing on edge optimisation. A lab implementation using LLaVa [88] on a Raspberry Pi 5 cluster has been accomplished separately [89]; distributed edge deployment is beyond the scope of this paper.

⁴This is often important in public safety use cases.
⁵The dataset is available at <https://github.com/Subramani-PRSM/Aerial-Detection-Faster-RCNN>

```

1 # Functional Definition for the Tools as per Specification in Listing 2
2 capture_image_func = FunctionDeclaration(name="capture_image", description="The act of capturing
    image from onboard camera.", parameters={"type": "object"})
3
4 object_func = FunctionDeclaration(name="object_image", description="The act of identifying objects in
    capture image using zero-shot open vocabulary model", parameters={"type": "object"})
5
6 segment_func = FunctionDeclaration(name="segment_image",description="The act of using the object
    definition to identify users and constructing the prompt for segmentation yielding mask for the
    identifying the users.", parameters={"type": "object"})
7
8 centroid_func = FunctionDeclaration(name="centroid_masks", description="The act of finding centroid
    of the user masks.",parameters={"type": "object"})
9
10 dist_func = FunctionDeclaration(name="dist_centroid", description="The act of calculating the
    distance between the current GPS location and centroid of the user mask cluster.", parameters={"
    type": "object"})
11
12 nav_func = FunctionDeclaration(name="nav_centroid",description="The act of going moving to the
    centroid of user cluster for serving them.", parameters={"type": "object"})
13
14 path_func = FunctionDeclaration(name="path_calc", description="The act of calculating the path-loss
    and downlink rates.", parameters={"type": "object"})
15
16 done_func = FunctionDeclaration(name="done", description="The goal has been reached", parameters={"
    type": "object" } )
17
18 uav_tools = Tool(function_declarations=[capture_image_func,object_func,segment_func,centroid_func,
    dist_func,nav_func,path_func,done_func ])

```

LISTING 3. Function specification.

2) CHOICE OF FOUNDATION MODELS FOR SOLVING THE TASK

For the agent implementation, we employ Gemini 1.5 [69] as the multimodal foundation model. For tracking tasks, we use Meta's Segment Anything Model 2 (SAM2) [90], which extends SAM by incorporating temporal dynamics for video segmentation through a Promptable Visual Segmentation (PVS) task. SAM2 takes input points, boxes, or masks on any frame to predict and track spatio-temporal masks (masklets). The key components are shown in Fig. 7.

Since SAM2 requires a prompt (bounding box or points) to create masklets, we invoke a zero-shot object detection model, Grounding DINO [91], to generate bounding boxes automatically. This combination—known as Grounded SAM [92] for the original SAM—provides precise user tracking for the UAV platform. In summary, our reference implementation combines Gemini 1.5 and SAM2,⁶ initialised as follows⁷:

⁶The recently released LLaVa-plus [93] provides an alternative with a single foundation model.

⁷We aim to publish the GitHub repository after publication. The configurations are similar to the example at https://github.com/GoogleCloudPlatform/generative-ai/blob/main/gemini/function-calling/intro_diy_react_agent.ipynb

3) TOOLSET AND FUNCTION SEMANTIC DEFINITIONS

The agent is furnished with functions that aid in solving its assigned tasks. Parameters can be extracted from the intent prompt or passed between functions via an invocation context. In our implementation, a global mission state tracks the agent's progress, captured images, and segmentation masks, accessible by all invoked functions. The toolset has two key elements:

a: FUNCTION IMPLEMENTATION

Eight core functions (see Fig. 8) are implemented alongside helper functions for visualisation and text formatting. The function `capture_image` captures an image (selecting from the dataset; in real UAVs, this would implement hardware-dependent capture via SPI/I2C and OpenCV preprocessing). The function `object_image` performs zero-shot object recognition to create bounding box prompts, and `segment_image` generates segmentation masks.

The function `centroid_mask` applies K -means clustering [94] to the masks to identify user groups, with centroids selected as candidate UAV positions that minimise path-loss. This extends to multi-UAV scenarios where agents coordinate user coverage. We determine K via the multimodal LLM by

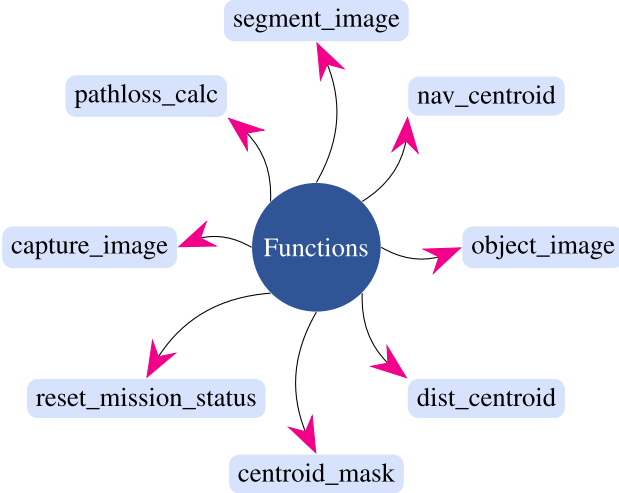


FIGURE 8. Functional structure for the agents.

querying the captured image to estimate the number of distinct user groups, which initialises the clustering algorithm. This shows that the LLM can be used within tool functions, not only for agent orchestration.

While K -means is efficient for semi-static scenarios, it assumes a fixed cluster count and centroid-based partitioning, which may not capture irregular or time-varying user distributions. Alternatives such as DBSCAN handle non-uniform distributions without requiring K a priori. In the proposed framework, such methods can be exposed as interchangeable tools in the function-calling pipeline, allowing the agent to select the most suitable strategy based on the observed environment. Comparative evaluation of these approaches in dynamic scenarios is left for future work.

b: FUNCTION DESCRIPTION/SEMANTIC INFORMATION FOR MODEL

Function descriptions are provided to the model following the Open API Specification 3.0:

Definitions of all implemented functions are provided in Listing 3.

Appendix A, Fig. 13 shows a sample execution from the implemented agent. Since GPS coordinates are not available in the aerial dataset, centroids in the image plane require inverse matrix transformation to locate positions in the x - y plane. To facilitate demonstration, we created a simulation environment in ThreeJS⁸ providing the capability to fly a drone in a virtual environment and capture images from the UAV perspective. Alternatively, AirSim [95] or similar simulators interfaced with flight controllers (e.g., PX4⁹) can be employed. Fig. 9 shows snapshots of the agent's operation on virtual images, producing output consistent with the actual scenario execution in Fig. 13.

⁸We use the React framework with the ThreeJS wrapper React3Fiber. <https://threejs.org/>

⁹Tutorial at https://docs.px4.io/main/en/sim_airsim/index.html

4) PERFORMANCE EVALUATION AND RESULTS

The purpose of this case study is to validate the end-to-end agentic reasoning pipeline and quantify its impact on network-level performance. The agent autonomously completed the coverage optimisation task in 7 reasoning cycles (as shown in Fig. 13), invoking 8 function calls without any irrelevant invocations, demonstrating correct relevance detection behaviour. The characteristics of the underlying foundation model—including latency, relevance detection, and energy consumption as formalised in Section II-B—further influence overall system efficiency. Gemini 1.5 was selected for its strong multimodal capabilities and function-calling accuracy (see Fig. 2); systematic cross-model comparisons using the metrics defined in Section II-B can be performed for deployment-specific model selection. A comprehensive cross-model benchmarking study—profiling latency, energy consumption, and relevance detection across multiple foundation models on identical telecom tasks—constitutes a separate research effort and is beyond the scope of this framework-oriented contribution. The metrics formalised in Section II-B are designed to serve as a standardised basis for such future evaluations.

To characterise the network-level performance gains, we introduce a mathematical framework for quantifying quality-of-service (QoS) metrics, focusing on two key measures: average throughput and outage probability. Average throughput incorporates line-of-sight (LoS) propagation by averaging over small-scale fading. For outage probability, we consider a desired QoS target rate R_{th} accounting for small-scale fading variations. Consider the UAV covering N distinct users with locations $\Pi = \{\mathbf{p}_i\}$, $i \in \{1, 2, \dots, N\}$, $\mathbf{p}_i = [x_i, y_i, z_i] \in \mathbb{R}^{3 \times 1}$, forming a spatial point process [96] in $\mathbb{R}^3$. Let $\mathbf{p}_o = (x_o, y_o, z_o)$ denote the initial UAV location. The channel between the UAV at $\mathbf{p}_j$ and user $\mathbf{u}_i \in \Pi$ is modelled as

$$h_{ij} = g_{ij}d_{ij}^{-\alpha/2}, \quad (3)$$

where h_{ij} is the overall channel gain decomposed into distance-dependent path-loss $d_{ij}^{-\alpha/2}$ and small-scale fading g_{ij} . The distance d_{ij} is

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}, \quad (4)$$

with path-loss exponent $\alpha = \alpha_o + \alpha_1(\theta_{ij})$, where θ_{ij} is the elevation angle. For dominant LoS scenarios, the elevation-angle dependence is negligible and $2 \leq \alpha_0 \leq 6$ suffices [97], [98]. We assume Rician fading [97], [98] with $\mathbb{E}(\|g_{ij}\|^2) = 1$ and

$$g_{ij} = \sqrt{\frac{K_{ij}}{K_{ij} + 1}}g + \sqrt{\frac{1}{K_{ij} + 1}}\bar{g}, \quad (5)$$

where g is the deterministic LoS component ($\|g\| = 1$), $\bar{g} \sim \mathcal{CN}(0, 1)$, and $K_{ij} = A_1 \exp(A_2\theta_{ij})$ is the Rician factor bounded by $K_{min} \leq K_{ij} \leq K_{max}$ [98]. The received SNR is

$$\Gamma_{ij} = \frac{\gamma P_T \|h_{ij}\|^2}{\sigma^2}, \quad (6)$$

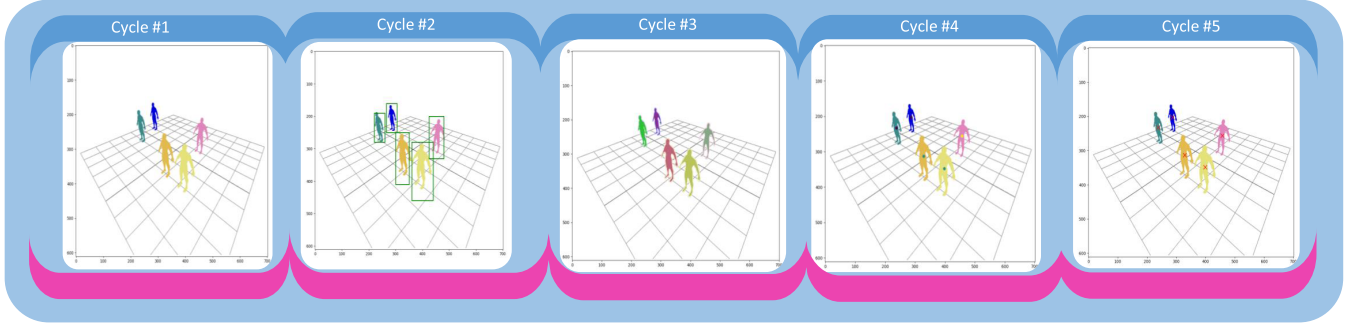


FIGURE 9. Simulated environment in three.js Aerial simulator.

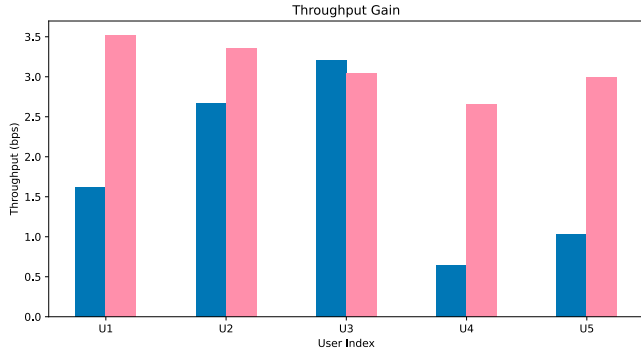


FIGURE 10. Average Throughput $\bar{R}$ for the users before (blue) and after (pink) optimal positioning of Multimodal Agent. The computation of the average throughput employs Eq. (8). $\frac{P_T}{\sigma^2} = 100\text{dBm}$, $\alpha = 4$, h_{uav} and h_{ue} are needed to calculate the 2D distance from 3D arena. The absolute values are not important but the ratio of these should be appropriate, e.g. $h_{uav} = 300\text{ ft}$ and $h_{ue} = 6\text{ ft}$ in our simulations correspond to $h_{uav} = 5.385$ and $h_{ue} = 0.1$, multiplying both with a factor of 60 provides correct scaling.

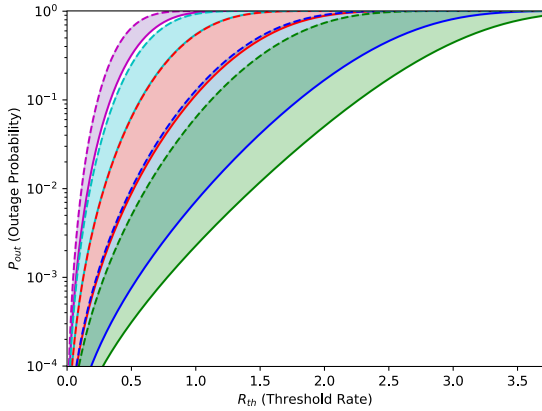


FIGURE 11. The outage probability for different users before (dashed curve) and after (solid curve) optimal repositioning of the Multimodal LLM Agent assuming $K_{ij} \leq K_{max} = 10\text{dB}$. The band colour signifies the gain. The outage probability is plotted using the Eq.(9) in the analysis.

with P_T the transmit power, σ^2 the noise power, and γ incorporating frequency, antenna, and hardware factors. The achievable rate is

$$R_{ij} = \log_2 (1 + \Gamma_{ij}) \text{ b/s/Hz.} \quad (7)$$

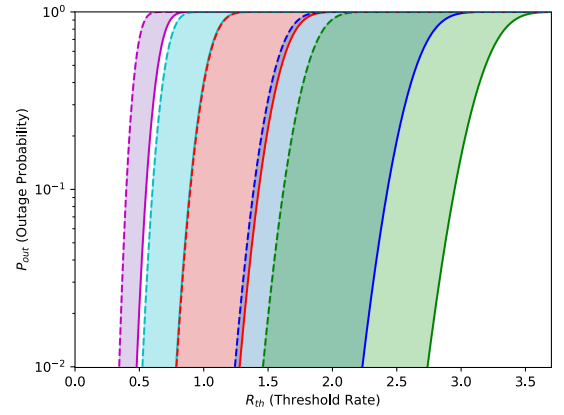


FIGURE 12. The outage probability for different users before (dashed curve) and after (solid curve) optimal repositioning of the Multimodal LLM Agent assuming $K_{ij} \leq K_{max} = 20\text{dB}$. The band colour signifies the gain. The outage probability is plotted using the Eq.(9) in the analysis.

Averaging over small-scale fading ($\mathbb{E}(g_{ij}) = 1$) yields the average throughput:

$$\bar{R}_{ij} = \log_2 \left(1 + \underbrace{\frac{\gamma P_T}{d_{ij}^{-\alpha} \sigma^2}}_{\bar{\Gamma}_{ij}} \right) \text{ b/s/Hz} \quad (8)$$

The outage probability for intended rate R_{th} is:

$$\begin{aligned} p_{out}^{i,j}(R_{th}) &= \mathbb{P}(R_{ij} < R_{th}), \\ &= F_{\|g_{ij}\|^2} \left(\frac{2^{R_{th}} - 1}{\bar{\Gamma}_{ij}} \right), \\ &= 1 - Q_1 \left(\sqrt{2K_{ij}}, \sqrt{2(K_{ij} + 1)(2^{R_{th}} - 1)\bar{\Gamma}_{ij}^{-1}} \right), \end{aligned} \quad (9)$$

where $\bar{\Gamma}_{ij} = \frac{\gamma P_T}{\sigma^2 d_{ij}^{-\alpha}}$ and $Q_1(a, b) = \int_b^\infty x \exp\left(-\frac{x^2 + a^2}{2}\right) I_0(ax) dx$ is the Marcum Q function [99]. Since $p_{out}^{i,j}(R_{th})$ depends on d_{ij} and R_{th} , the agent's performance after repositioning is evaluated by setting $\mathbf{p}_j$ to the final UAV location.

5) RESULTS

Fig. 10 demonstrates the change in UE throughput after the UAV repositions to complete the specified intent.



FIGURE 13. Execution of the multimodal agent for the specified intent and goal.

For most UEs, throughput increases after repositioning to the cluster centroid, though some UEs may experience decreased throughput due to increased path-loss. Only UE3’s performance deteriorates slightly. The pre-repositioning

performance serves as the baseline, representing operation from the initial unoptimised UAV position. The gains reported in Figs. 10–12 therefore quantify the value added by the autonomous agentic pipeline over static deployment. The goal is system-level optimisation; sacrificing individual throughput for a few UEs is an acceptable trade-off. The proposed framework is general and supports more complex placement and trajectory optimisation [100], including heterogeneous demand parameters and service class types (eMBB, URLLC, etc.).

Fig. 11 plots outage probability versus rate threshold R_{th} with Rician factor $K = 10$ dB. Outage probability reduces for all UEs after repositioning, with the extent of reduction varying per UE (indicated by coloured bands). For UE5 (green band), the reduction is significantly larger than for UE1 (magenta), consistent with the throughput gains in Fig. 10.

Fig. 12 plots outage probability with $K = 20$ dB. As expected, the stronger LoS component improves outage performance for all UEs compared to Fig. 11, with steeper outage curves indicating support for higher rates at low outage thresholds.

V. FUTURE WORK AND RESEARCH DIRECTIONS

While this article validated the proposed framework through a UAV coverage optimisation case study, several directions remain open:

- 1) **Probabilistic reasoning for function invocation:** Wireless networks are dynamic environments where multiple actions can yield similar outcomes at different costs. While this paper formalises stochastic function calling to generalise ReAct, further exploration—including probabilistic reasoning [101] for action selection and its application to wireless use cases—remains future work.
- 2) **Multimodal RAG integration:** Expanding beyond text–vision fusion to include time-series and RF features in a unified embedding space could enhance multimodal reasoning. Whether these should be embedded directly or handled by a dedicated foundation model with task-specific layers requires further investigation for 6G RRM use cases such as automated RAN slicing.
- 3) **Edge–cloud continuum:** The framework has been validated in a cloud environment. Extending this to distributed inference across the edge–cloud continuum with hardware-aware optimisations (e.g., quantisation, model partitioning) remains an open challenge [102], particularly for multimodal agents with in-context learning.
- 4) **Personas and safety:** Systematic evaluation of how persona configurations affect reasoning trajectories and outcomes is needed. Developing measurable safety metrics, transparency tools for function invocation chains, and explicit bounds on generative

behaviour is essential for stability in real-time telecom deployments.¹⁰

- 5) **Other 6G use cases:** Applying the framework to network slicing orchestration, beam management, or non-terrestrial network handovers can reveal use-case-specific requirements for agentic implementation.

VI. RISKS AND LIMITATIONS OF AGENTIC AI IN TELECOM

Deploying LLM-based agents in telecom networks introduces several challenges that must be addressed for practical viability.

A. HALLUCINATIONS AND INCORRECT REASONING

LLM-based agents may generate incorrect reasoning steps or invoke inappropriate functions, potentially leading to harmful network control decisions. Mitigation strategies include strict tool-use constraints, output verification mechanisms, and the relevance detection metrics formalised in Section II-B.

B. ADVERSARIAL ROBUSTNESS

Since agent behaviour is driven by prompts and contextual information, malicious or ambiguous inputs could manipulate decision-making. This necessitates guardrails, input validation, and policy enforcement layers within the agentic pipeline [73].

C. PRIVACY AND DATA SECURITY

Handling sensitive data—including user locations, network usage patterns, and multimodal inputs (visual and RF)—raises significant privacy concerns. On-device processing, data anonymisation, and secure communication protocols are essential for regulatory compliance.

D. COMPUTATIONAL OVERHEAD

Large-scale LLM deployment may limit real-time applicability in resource-constrained environments. Edge–cloud deployment strategies can partially address this, but careful system design is required to balance latency, energy consumption, and performance.

Addressing these challenges is critical for safe deployment of agentic AI in future 6G networks and remains an important research direction.

VII. CONCLUSION

This paper presented a comprehensive framework for multimodal ReAct-based agents to enable intent-driven management in 6G wireless networks. By integrating reasoning, acting, and multimodal data fusion, the proposed architecture enables autonomous orchestration of complex tasks. The framework combines function calling, persona-driven reasoning, and multimodal retrieval-augmented generation to translate user intent into actionable network optimisations. The practical viability of the framework was demonstrated

¹⁰This area is also in its infancy: <https://www.aisi.gov.uk/work/strengthening-ai-resilience>

through a UAV-based coverage optimisation case study, in which the agent autonomously completed the task through seven reasoning cycles, yielding substantial reductions in outage probability and improvements in throughput under Rician fading channels. The case study provides design guidelines for similar implementations in other 6G use cases, paving the way for a new generation of self-optimising networks.

APPENDIX A SAMPLE EXECUTION FOR THE AGENT

The sample execution is provided in Fig. 13.

APPENDIX B TRAD FRAMEWORK FOR AGENT (FROM [68])

Require: o_t : Observation at timestep t .

π_{LLM} : Pre-trained language model.

M : Thought-enhanced memory with expert trajectories.

Hyperparameters B, F, K : Number of backward steps, forward steps, and top-K retrieved step.

a_t : Action at timestep t .

Step 1: Thought Generation

- 1: Generate a thought τ_t for the current state using the LLM:

$$\tau_t = \pi_{LLM}(o_{0:t}, a_{0:t-1}).$$

Step 2:Thought Retrieval

- 2: Perform a dense similarity search to retrieve relevant demonstration steps:
- 3: Encode τ_t as a query vector.
- 4: Search M to find the top- K steps $\{s_1, s_2, \dots, s_K\}$ with the highest similarity scores to τ_t .

Step 3:Temporal Expansion

- 5: **for** each retrieved step s_k **do**
- 6 Include B previous and F subsequent steps:

$$\text{Seq}_k = [s_k^{-B}, \dots, s_k, \dots, s_k^{+F}]$$

7: **end for**

Step 4:Relative Order Marking

- 8: **for** each step in Seq_k **do**
- 9 Label each step with its relative temporal position (e.g., [Step -1], [Step 0], [Step +1]).

10: **end for**

Step 5:History Alignment

- 11: Combine the current observation o_t with up to B previous observation-action pairs:

$$\text{History-Aligned Input} = [o_{t-B:t}, a_{t-B:t-1}].$$

Step 6:Action Prediction

- 12: Input the enriched context (retrieved sequences + current state) into the LLM for decision-making:

$$a_t = \pi_{LLM}([\text{History-Aligned Input}; \text{Seq}_1, \dots, \text{Seq}_K]).$$

Step 7:Output Action

- 13: Execute the predicted action a_t . = 0

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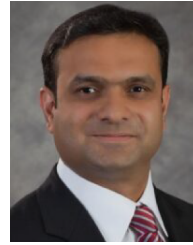
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